CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

University Examination

Date: 27/11/2019 **Day: Wednesday** **Time: 10:00 am to 10:30 am** **Total Marks: 20**

Student's ID No.	Answer Sheet for Multiple Choice Questions (MCQs),True or false and Match the following
	Examination : BOP - November 2019
Student's Name: ____________	Semester: II
Student's Sign : ____________	Section : I
Supervisor's Name : ____________	Course Code : BOP 204
Supervisor's Sign: ____________	Course Name: Physical Optics

Important Instructions:

1. There are two sections in the paper.
2. **SECTION –I** contains 2 Questions:
   a. MCQs for 20 marks answered in the question paper itself, should be submitted within 30 minutes.
   b. Short answers for 20 marks (to be answered in separate answer book)
3. **SECTION – II** contains only 2 Question of Short notes and Essays (to be answered in answer book)
4. Marks on the right side suggest the total marks of that question.
5. Maximum time allotted for both section is 3 hours.
6. Candidates who complete **SECTION – I** (MCQs) earlier than 30 minutes can collect main question paper immediately after handing over the **SECTION-I (MCQs)** question paper with answers.
7. Non-programmable calculators are only allowed.
8. Draw the figures where necessary.

SECTION-I

Q.1. Multiple Choice Questions: (MCQs) **(10 × 1 = 10)**
(Encircle the correct answer)

1. **Which of the relation is correct for SHM,**
   a. $a \propto x^2$
   b. $a \propto -x$
   c. $a \propto \sqrt{x}$
   d. $a \propto x$
2. **In terms of wavelength, below which relation is correct,**
   a. Infrared < Visible < Gamma
   b. X-ray >UV-ray > Microwave
   c. Visible > X-ray < UV-ray
   d. Gamma > X-ray < Infrared

3. **Polaroid readily absorbs light whose electric field vector is ………….. to the transmission axis.**
   a. Parallel
   b. Quarter
   c. Perpendicular
   d. None of the above
4. **Polarized light has………intensity of the unpolarized light.**
   a. Half
   b. Double
   c. Quarter
   d. Triple
5. **At which angle between polarized and analyzer the passing wave has a lowest intensity and magnitude?**
   a. 20
   b. 30
   c. 45
   d. 60
6. **Fringe width β is independent of …………….**
   a. Wavelength of the light
   b. Distance between screen and slits
   c. Distance between two slits
   d. Order of fringe
7. **If the optical path difference between two waves contains an integral number of wavelength then ………will occurs on screen.**
   a. Dead Spot
   b. Bright fringe
   c. Dark fringe
   d. None of the above
8. **In Rayleigh scattering, light of ………..scattered more.**
   a. 470 nm
   b. 520 nm
   c. 580 nm
   d. 670 nm
9. **In fluorescence, the emitted light has ……….. wavelength then that of absorb light.**
   a. Shorter
   b. Longer
   c. Same
   d. None of the above
10. **Which process is independent of incident photon intensity…………..**
   a. Absorption
   b. Stimulated emission
   c. Spontaneous emission
   d. None of the above

Identify the sentence as True or False: **(5 × 1 = 5)**
(Please tick mark the answer) True False

11. All the oscillatory motions are periodic motions.
12. Brewster angle depends on refractive index of the reflecting surface.
13. For interference, two light waves must be coherent.
14. Fluorescence is faster than phosphorescence.
15. The light emitted by spontaneous emission is coherent light.

Match the following **(5 × 1 = 5)**

16. Christian Huygens **17. Sir Isaac Newton** **18. Thomas Young** **19. Malus** **20. Rayleigh**		**A. Interference** **B. Resolution** **C. Wave Theory** **D. Fluorescence** **E. Polarization** **F. Particle Theory** **G. Scattering**

(16 -), (17-), (18-), (19-), (20-)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of BOP Examination

November 2019

BOP 204: Physical Optics

Date: 27/11/2019 Day: Wednesday Time: 10:30am to 1:00pm Maximum Marks: 50

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SECTION-I

Q.2. Short answers (10 out of 11) **(10 × 2 = 20)**

a. Define periodic and oscillatory motion.

b. Define transverse and longitudinal nature of the wave.

c. Define Frequency, Wavelength, Period and Amplitude.

d. Differentiate between polarized and un-polarized light. (2 points each)

e. Define coherent and incoherent wave.

f. Define constructive and destructive interference.

g. Define resolution. State the Rayleigh's criterion for resolution.

h. Define and derive the equation of Brewter's angle.

i. Define solid angle with its equation.

j. State Inverse square law with its equation.

k. Define birefringence with a clean diagram.

SECTION-II

Q.1. Short Notes (4 out of 5) **(4 × 5 = 20)**

a. Write a short note on (i) Visible light, (ii) Ultraviolet light and (iii) X-rays.

b. Explain plane and circularly polarized light.

c. Derive the equation for resolving power of telescope.

d. Explain Phosphoresce phenomenon.

e. Define different radiometric units.

Q.2. Essay (1 out of 2) **($1 \times 10 = 10$)**

a. Explain Young's double slit experiment. Derive the equation for optical path difference and state the conditions for dark and bright fringe.

b. Explain the emission of laser by Stimulated emission and also write its characteristics.